

Visualizing Climate Model Outputs with CliMAScope

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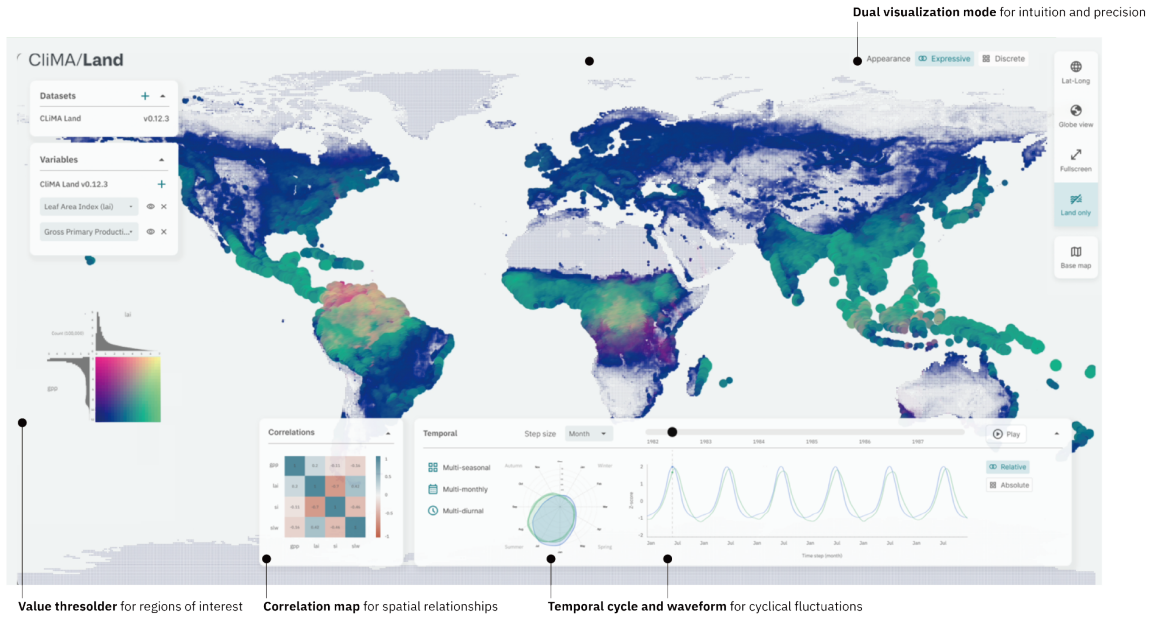


Figure 1: The *CliMAScope* interface combines multiple visualizations to understand spatial, temporal, and inter-variable relationship patterns for climate data.

ABSTRACT

With the increasing complexity and diversity of climate models, climate scientists often turn to multiple tools to understand and assess relationships between observations and model output. To explore ways to overcome that workflow complexity, we partnered with the Climate Modeling Alliance (CliMA) to develop *CliMAScope*. *CliMAScope* is an open source, interactive, web-based land surface climate model visualization tool that aims to unify analysis, comparison, validation, and correlation mapping of observational data with climate land model output. This paper presents a design study of *CliMAScope*, and explains how the tool's visual and interactive elements support the goal of comparing seamless climate model outputs to observational data.

Index Terms: visual analytics, design study climate model

1 INTRODUCTION

The Climate Modeling Alliance (CliMA) is a coalition of climate scientists, engineers and mathematicians working to create a more accurate, open source, and modular climate model [26, 6]. One of CliMA's goals is to develop new tools to understand model outputs and assessing model performance against empirical data and alternative implementations.

As climate models grow in size, complexity, and diversity of components, assessing model accuracy becomes increasingly complex [11]. Climate model researchers use various, often static visualizations to interpret model outputs and compare them with other models [25] as part of their iterative scientific analysis and development. Because analyses require making connections between different features, time steps, and data sources, multiple figures are often required. However, static figures do not allow for linking, brushing, and filtering that allow climate scientists to identify underlying patterns.

To address these challenges, we partnered with the CliMA science team and developed *CliMAScope*, an open source, web-based tool that supports flexible and consistent analysis of CliMA model data [19]. *CliMAScope* includes three core functions: comparison, temporal analysis, and contextual highlighting. This paper presents a design study [23] of the *CliMAScope* visual analytics application which explains the data analysis and workflow of climate scientists and how *CliMAScope* aims to support that analytical process.

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2 BACKGROUND

2.1 Investigating Earth Systems and Climate Models

Earth system models (ESMs) are fundamental tools for climate science research, simulating complex interactions among the Earth's atmosphere, oceans, land surfaces, and biosphere [10]. These models allow researchers to study critical processes such as carbon cycling, energy exchange, atmospheric dynamics, and greenhouse gas impacts across various spatial and temporal scales. Typical ESM outputs include temperature [16], precipitation [24], wind [21], and fluxes between the surface and atmosphere, such as evapotranspiration [18] and carbon fluxes through plant canopies [14]. They are typically produced at spatial resolutions ranging from tens to hundreds of kilometers and across timescales from weeks to centuries.

While traditional ESMs have significantly advanced our understanding of climate dynamics, many legacy modeling frameworks suffer from limitations related to computational efficiency, scalability, and modularity [3]. To address these challenges, the Climate Modeling Alliance (CliMA) has focused on developing a modern, modular ESM framework from scratch, leveraging open-source computing environments and novel computational approaches [26, 6, 25]. CliMAScope is designed to address general ESMs visualization challenges, alongside the improvements brought by CliMA.

2.2 Visual Analytics for Climate Modeling Research

Visual analytics play a crucial role in interpreting ESM outputs, enabling researchers to explore complex and high-dimensional climate data. Historically, visual representations have included static maps, time series plots, and simple animations (e.g., [10]), facilitating the basic interpretation and validation of the output of the climate model against observational datasets. However, as the complexity and dimensionality of ESM outputs have grown, traditional visualization methods have increasingly struggled to keep up with the data size and complexity, as well as provide effective and intuitive communication of dynamic multidimensional data.

Several research projects have investigated some of the visual encoding and interaction problems that scientists face when trying to understand ESMs. For example, at baseline, ESMs often include dense parametric data displayed in 2D geography [9]. Those maps, alongside commercial software [7], allow scientists and engineers to compare hypotheses [2] and tune climate model performance [11]. Other approaches have investigated rendering problems associated with high-throughput computation of atmospheric climate data [15], and clustered performance metrics across climate model similarity spaces [20].

CliMAScope broadly investigates visualization approaches that combine interactive capabilities to assess the usability, interpretability, and accessibility of ESM outputs [19]. We see this represents a novel, underexplored area. We argue that such approaches potentially enable researchers to ask new scientific questions, facilitate more effective hypothesis generation, and support deeper understanding of complex climate system dynamics.

3 DESIGN STUDY

This paper reports on a design study [23] of climate scientists and the *CliMAScope* application designed to support their analysis and investigation of climate models. As part of the Data to Discovery program [8], the design team spent 10 weeks, approximately 40 hours each week, engaged with the science team. We began with contextual inquiry [4], observing the scientists' work processes in two-hour sessions. Through these sessions we identified potential friction points and breakdowns in their existing process and developed an understanding of the users, their data, and their workflows. During the next several weeks, the design and science teams used lightweight prototyping [1] to co-design possible visual encodings and interactions [22]. The final phase of the design study included weekly think-aloud evaluations of more mature prototypes [17],

and co-design of future features. The software prototype described later in this paper was iteratively developed in Javascript using D3 [5].

4 USER WORKFLOW

Climate researchers rely on visualizations that help answer key questions about their datasets and outputs, including:

- Q1 How accurately do models reflect real-world climate data through residuals?** Researchers analyze the residuals between model predictions and actual observations to identify discrepancies, track errors, and validate model assumptions. Interactive tools such as ~~dynamic timelines and spatial overlays~~ visualize these residuals and support the evaluation of model accuracy against empirical benchmarks.
- Q2 What temporal dynamics emerge in climate systems?** Scientists explore how climate variables evolve over time, identifying recurring fluctuations and rare anomalies in temporal graphs and climate spirals to better understand short-term variability and long-term shifts in environmental conditions.
- Q3 How do key climate variables vary spatially?** Scientists use interactive global and regional maps to identify where and how climate variables, such as Gross Primary Production, evapotranspiration, and soil moisture, differ across locations.
- Q4 How do climate variables influence each other in time and space?** Beyond individual variable changes, scientists also investigate how these variables interact. Using visualizations, they track how a shift in one variable, like a temperature anomaly, can influence others such as soil moisture or evapotranspiration across regions and time frames.

5 DATA

The datasets used in *CliMAScope* come from the Climate Modeling Alliance (CliMA) Land model outputs. They provide high-resolution climate simulations designed to improve our understanding of Earth system dynamics. In our design process, we utilized the CliMA Land dataset version 0.12.3, which includes monthly data from 1982 to 1987 at a resolution of ~40 - 80km. This dataset offers insights into over 60 key climate variables, enabling analysis of long-term patterns and variability. The dataset is structured over multiple spatial and temporal resolutions. The global grid-based data allows for visualization of output variable distributions across different regions, while the temporal component enables users to track seasonal and interannual variations. The current implementation supports this baseline spatial and temporal resolution of standard, extant modern climate models.

In addition to the standard model output, the dataset includes derived statistical metrics and spatial correlation maps, which help to identify patterns across different latitudes, seasons, and ecological zones. To further enhance our analysis, elevation data was incorporated, enabling users to assess how terrain influences vegetation and productivity.

5.1 The *CliMAScope* Application

CliMAScope provides an interactive and multifaceted environment for climate scientists to explore, analyze, and compare climate model outputs. The interface consists of several key views that guide users through different analytical workflows, facilitating intuitive exploration of climate data.

5.1.1 Spatial Understanding

At the core of *CliMAScope* is the **geospatial visualization panel**, which supports spatial exploration (Q1, Q3). Upon uploading datasets (e.g., the CliMA Land dataset), users can select climate

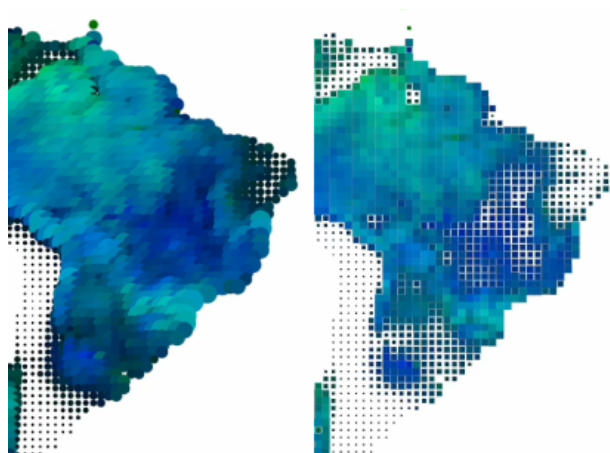


Figure 2: A comparison of the expressive (left) and discrete (right) visual styles used by *CliMAScope*. The expressive visual style exaggerates the spatial data, using overlapping circles to create an organic appearance, while the discrete visual style limits the size of the square to the size of the geospatial pixel.

variables such as Leaf Area Index (LAI) or Gross Primary Productivity (GPP). These variables are then rendered as continuous heatmaps on a global map, with optional grid lines to highlight latitudinal and longitudinal gradients.

To accommodate different modes of reasoning, the panel includes two visualization styles (Figure 2): (1) **Expressive mode** uses painterly overlapping circles [12, 13] that exceed the bounds of their geospatial pixel, producing a slightly exaggerated, impressionistic rendering. This non-standard encoding emerged through iterative co-design with scientists and mentors, who noted that anomalies often stood out in this view—drawing attention viscerally and encouraging ambient pattern recognition before closer inspection. The expressive mode supports holistic interpretation through spatial gestalt rather than individual marker size, allowing broad spatial patterns to remain perceptible even when rendering order introduces minor visual variation. (2) **Discrete mode** renders geospatial pixel-level precision using bounded square pixels, enabling detailed interrogation of anomalies or boundary conditions in the data. In both representations, the size and color of each datapoint represent the value of the location relative to the minimum and maximum values for the variable. This dual-mode approach supports fluid transitions between overview and detail: users can begin in expressive mode to scan for emergent patterns, then switch to discrete mode for targeted inspection.

In tandem, **value thresholding** (Figure 3) allow users to isolate regions of interest (e.g., high GPP in the Amazon) and define thresholds in interactive histograms, a 2D color map and a corresponding spatial map. Rectangular brushes in the histograms and color map region filter values of interest. The corresponding map dynamically updates to highlight only regions within the selected GPP and LAI thresholds, enabling targeted exploration of climate-vegetation relationships. **Topographic overlays** (Figure 6) provide contextual layers (e.g., elevation) to investigate how terrain influences climate variables. A full screen mode removes interface distractions for immersive analysis.

5.1.2 Temporal Understanding

To investigate temporal dynamics (Q2, Q4), *CliMAScope* includes a **temporal panel** located below the map. This panel features two primary visualizations (Figure 4): (1) an **aggregate view**, showing seasonal, monthly, or diurnal summaries, and (2) a **temporal wave-**

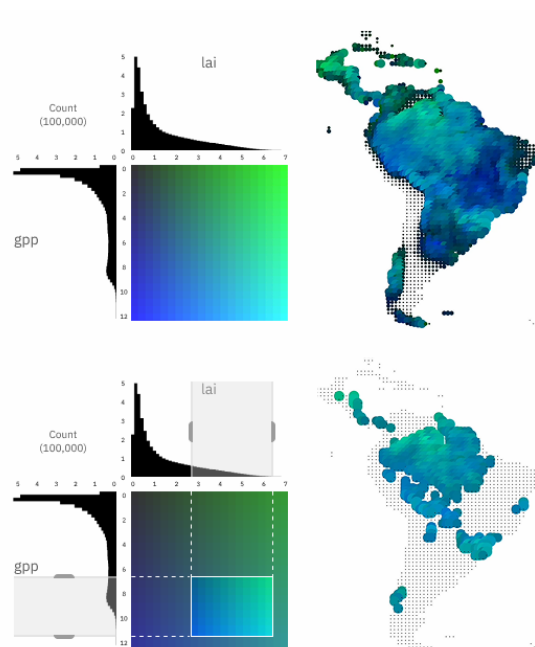


Figure 3: *CliMAScope*'s interactive thresholding view. Users brush value ranges for GPP and LAI on linked histograms and the bivariate color map (left). The spatial view (right) updates accordingly. Two examples of enabled comparisons are the full bivariate distribution of GPP and LAI over South America (top row) and a user-defined selection (bottom row).

form and radial plot, revealing cyclical fluctuations across days, months, or years.

The radial plot aggregates values across all years into a circular layout, mapping months to angles. This representation highlights seasonal patterns and cyclicity. In the example shown, LAI (green) and GPP (blue) exhibit strong shape alignment but differ in phase: the GPP curve consistently peaks earlier than LAI throughout the seasonal cycle, especially around May to July. This visual offset supports hypotheses around ecological lag, such as photosynthetic activity (GPP) responding to light and temperature conditions before structural vegetation growth (LAI) follows.

The temporal waveform chart on the right visualizes continuous time-series data across multiple years. Each line encodes one variable, enabling direct comparison of amplitude, periodicity, and interannual variation. The consistent lead of GPP over LAI in

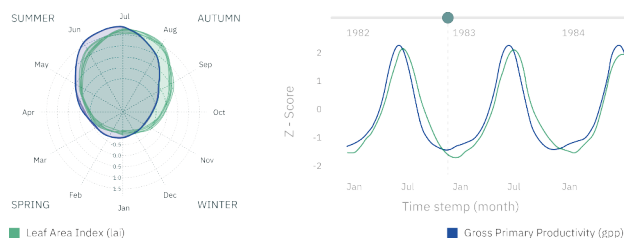


Figure 4: The temporal panel, located at the bottom of the *CliMAScope* interface includes an aggregate view (left) depicting patterns across several timescales, and a temporal waveform (right) showing the timeseries data.

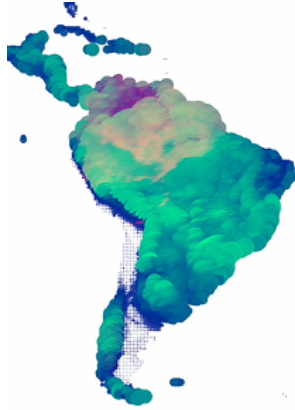


Figure 5: A combined GPP-LAI view reveals unexpected color shifts over the Amazon basin, suggesting anomalous model behavior.

each annual cycle becomes visually apparent as a horizontal offset between wave peaks. Such patterns may suggest lead-lag relationships critical for modeling carbon fluxes or validating satellite-derived vegetation indices.

These plots are tightly linked to the selected variable(s), allowing users to explore time-based patterns. For instance, in the geospatial visualization panel, an animated view of LAI changes over months reveals seasonal shifts in vegetation across latitudes. In tandem, analysis of time lags between variables, when peaks and troughs are tightly coupled but temporally offset, is made possible by comparative inspection of the overlapping peaks and dips of the GPP waveforms (Figure 4, right side, blue) offset from those of the LAI waveform (Figure 4, right side, green), prompting new hypotheses about ecological behavior.

5.1.3 Inter-Variable Understanding

To support multi-variable reasoning (Q4), a **correlation map** in the lower left visualizes spatial relationships between selected variables. For instance, when LAI and GPP are overlaid, a combined color encoding highlights areas of strong or weak correlation.

This panel is synchronized with the geospatial view, allowing users to explore correlation breakdowns across regions (e.g., high GPP but low LAI in the Sahel). Such insights support deeper ecological inference and model validation.

5.1.4 Understanding Model Performance

CliMAScope supports model evaluation by enabling direct comparison across datasets or model versions (Q1). When a second dataset is loaded, all views—spatial, temporal, and inter-variable—transition to show differences or deviations, rather than absolute values.

Beyond comparison, *CliMAScope* is designed to show unexpected behaviors in individual models. For example, in the combined GPP-LAI view (Figure 5), an anomalous color change over ecologically stable regions can quickly signal a misrepresentation in the model output. These visual cues, grounded in domain knowledge, allow scientists to spot implausible patterns, triggering further inspection or refinement. This workflow enables researchers to detect systematic biases, evaluate ecological coherence, and conduct visual hypothesis testing across space and time.

5.1.5 Contextualization

To support situated interpretation, *CliMAScope* includes overlays such as elevation, land cover, and political boundaries. These static or semi-interactive layers provide contextual grounding for

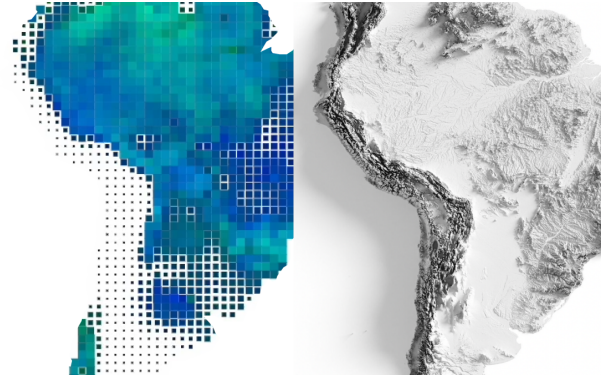


Figure 6: *CliMAScope* supports the integration of contextual layers, such as elevation data (right), alongside climate variables (left).

observed patterns in the data. For example, Figure 6 shows a side-by-side view of GPP-LAI patterns and the elevation map of South America. Scientists can use this spatial context to explain anomalies such as low productivity in high-altitude regions or to refine hypotheses about biome distributions, water availability, or model misalignment. By situating climate data within recognizable environmental features, *CliMAScope* bridges abstract model outputs with real-world geography, enhancing the interpretability of scientific insights.

6 EVALUATION

CliMAScope was iteratively evaluated during each of our weekly usability tests. One scientist captured its broader value: “We are dealing with complex data and complex processes... The diagrams we work with are very hard for scientists to parse and nearly impossible for the rest of the world.” They described the tool as opening doors in their analysis that were previously blocked: “I think there’s true visual genius to finding simple, clear ways of representing relations among variables and developing these tools. It’s impactful both for making the science more effective but also for communicating the science more broadly.”

7 CONCLUSION

As climate models grow in complexity, effective tools for visualizing and interpreting Earth System Model outputs become increasingly essential. *CliMAScope* demonstrates how interactive visual analytics can bridge the gap between raw model outputs and meaningful scientific insights, providing climate researchers with an intuitive, exploratory interface.

By integrating spatial, temporal, and correlation-based visualizations, *CliMAScope* enables researchers to examine key climate variables such as Leaf Area Index (LAI) and Gross Primary Productivity (GPP) across multiple scales. Features such as dynamic overlays, temporal animations, correlation mapping, and interactive filtering facilitate vanguard exploration of climate patterns, supporting hypothesis generation and model validation. The ability to compare model outputs with observational data further enhances researchers’ confidence in model accuracy and performance.

Our findings underscore the importance of context-aware visualization techniques in climate research. The integration of expressive vs. discrete visualization modes, topographic overlays, and interactive benchmarks offers new ways to interpret climate interactions, revealing subtle dependencies between variables that might otherwise be overlooked in static visualizations. Additionally, our work highlights the potential for scalable, open-source frameworks to enhance climate modeling efforts, supporting the broader research community.

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